Markov Spring 2026 Homework 5

Due Mar. 9 on Gradescope

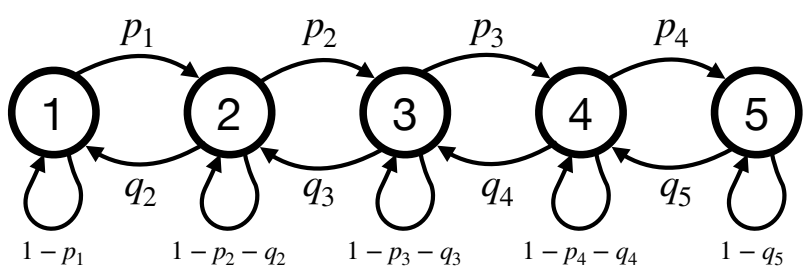
Plotting and reporting.

- **Document.** Submit a single document that includes all your figures.
- **Figures.** Include clean, labeled plots (axes, units, parameters) with consistent limits across panels. No photos of screens are accepted, although clear screenshots embedded in or appended to your document are Ok. Calculate key results such as empirical means, variances, and probabilities with 3–4 significant digits.
- **Code.** You don't need to include code in your written homework. Instead, create a github account if you don't already have one, create a public repository labeled "Markov2026", and upload your code there, ready to run. Label the code as "HW1_problem4.m", for example. Include the name of your account and repository in the homework. You can use any programming language you want (excel is strongly discouraged, though).
- **Math.** Accompany your calculations with clear and complete sentences. Discuss your results.

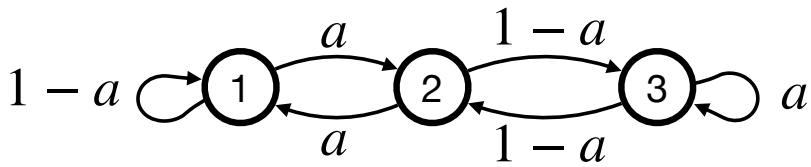
Problems

1. **Eigenvalues of Stochastic Matrices.** Suppose that p is a stochastic matrix.
   (a) Show that p has at least one eigenvalue equal to 1.
   (b) Show that all the other eigenvalues have magnitude less than or equal to 1. (Hint: you can use the Gershgorin circle theorem).
   (c) Show that if p_1 and p_2 are stochastic matrices of the same size, $p_1 p_2$ is also stochastic.

2. **Three versions of the Stationary Distribution.** Consider the Markov Chain with states $S = \{1, 2, 3, 4, 5\}$ and transition probabilities $p(n, n+1) = p_n = Ke^{an}$, $p(n, n-1) = q_n = Ke^{b(n-1)}$, $p(n, n) = 1 - p_n - q_n$, and $p(n, m) = 0$ if $|n - m| > 1$, as shown in the diagram below.



   (a) Use the detailed balance condition to find the stationary distribution $\vec{\pi}$. Find an expression for $\pi(i)$ and simplify it as much as possible.
   (b) Find the stationary distribution $\vec{\pi}$ by numerically finding the eigenvector of p^T corresponding to the eigenvalue 1 (don't forget to normalize it), using $a = 0.04$, $b = 0.16$, $K = 0.1$.
   (c) Simulate the Markov Chain using $a = 0.04$, $b = 0.16$, $K = 0.1$. for at least 10^6 steps. Plot a histogram of the fraction of time that the Markov Chain spent in the different states.
   (d) Plot your histogram from (c) together with the theoretical distributions found in (a) and (b). If your three histograms are not matching, there is something wrong somewhere.



3. **Convergence to the stationary distribution.** Consider the Markov Chain shown above, where $0 < a < 1$.

   (a) Find the stationary distribution π.

   (b) Find the eigenvectors and eigenvalues of the transpose of the probability transition matrix $A = p^T$. You are not only allowed, but encouraged, to use software to find the eigenvectors and eigenvalues of p as a function of a.

   (c) Set $a = 0.99$. Suppose that $X_0 = 1$. By decomposing $\mathbf{q}_0^T$ into eigenvectors of A and using $\mathbf{q}_{n+1}^T = A\mathbf{q}_n^T$, find an expression for $\mathbf{q}_n$ and show that it converges exponentially to π. For this part, you can work numerically and use software to do any required calculation.

   (d) For what value of a is the convergence to the stationary distribution faster? Find the value using your expression in (c) and also give an intuitive explanation.

   (e) Run $N = 100$ versions of the Markov Chain with $a = 0.99$, all of them starting at state 1. At time n, calculate the fraction f_n of the Chains that are in state 1. Plot f_n and the theoretical expression you found for $q_n(1)$ as a function of n on the same plot, for n ranging from 0 to a few hundred. Repeat for $N = 1000$ and $N = 10,000$. If they are not agreeing reasonably well, something is wrong!

4. **Proof of detailed balance for birth-death chains.** Consider a birth-death Markov Chain with state space $S = \{0, 1, 2, \ldots, N\}$ and transition probabilities $p(i, i+1) = p_i$, $p(i, i-1) = q_i$, $p(i, i) = 1 - p_i - q_i$, and $p(i, j) = 0$ if $|i - j| > 1$. Suppose that π is the stationary distribution. To prove that detailed balance applies between states i and $i+1$, define $P_i = \sum_{j=0}^{i} \pi_j$, and prove by induction that

$$P_i = P_i + q_{i+1}\pi_{i+1} - p_i\pi_i \tag{1}$$

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